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System and method for providing a bone cerclage

Abstract

A system for providing a bone cerclage according to an exemplary aspect of the present disclosure includes, among other things, a suture passer including a shaft. The shaft includes a passageway and a slot leading to the passageway. The slot has a variable width along a length of the passageway. A method is also disclosed.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to a system and method for providing a bone cerclage.

BACKGROUND

(2) Cerclages are used to maintain the relative position of fractured bone fragments or sections of cut bone to promote healing. While cerclages can be used relative to any fractured or cut bone, they

are commonly used to promote healing of femoral fractures. Cerclages are also used relative to other bones such as the humerus and sternum, as examples. In the context of the sternum, the sternum is cut during some surgical procedures, such as during open heart surgery. After the procedure, a cerclage may be used to maintain the relative position of cut sections of the sternum to promote healing. Some known cerclages include materials such as cable (i.e., wire) or fabric, and are provided by wrapping the cable or fabric around a fractured or cut bone a single time. After wrapping, the cable or fabric is then locked into place relative to the bone using a plug or clip, as examples.

SUMMARY

(3) A system for providing a bone cerclage according to an exemplary aspect of the present disclosure includes, among other things, a suture passer including a shaft. The shaft includes a passageway and a slot leading to the passageway, and the slot has a variable width along a length of the passageway.

(4) In a further embodiment, the system includes a sheath and a strand of suture projecting from the sheath.

(5) In a further embodiment, the sheath is more rigid than the strand of suture.

(6) In a further embodiment, the sheath exhibits a larger diameter than the strand of suture.

(7) In a further embodiment, the strand of suture is a first strand of suture and at least one additional strand of suture projects from the sheath.

(8) In a further embodiment, a length of the sheath is longer than a length of the passageway.

(9) In a further embodiment, the slot includes a first width adjacent an entrance of the passageway, the slot includes a second width along a main portion the passageway, and the first width is greater than the second width.

(10) In a further embodiment, the first width is sized so as to permit the sheath and suture to enter the passageway via the slot, and the second width is sized so as to prevent the sheath from exiting the passageway via the slot and so as to permit suture to exit the passageway via the slot.

(11) In a further embodiment, the slot is provided by a gap in a superior surface of the shaft.

(12) In a further embodiment, the shaft includes a curved section, and the passageway and slot are provided at least partially in the curved section.

(13) In a further embodiment, the suture passer includes a handle connected to the shaft, and the shaft includes a substantially straight section between the handle and the curved section.

(14) A method for providing a bone cerclage according to an exemplary aspect of the present disclosure includes, among other things, passing a sheath and a strand of suture around a bone. The sheath is connected to the strand of suture. Further, the method includes tensioning the strand of suture to tighten the strand of suture relative to the bone, and securing the strand of suture to the bone.

(15) In a further embodiment, the passing step includes positioning a suture passer relative to the bone, the suture passer includes a shaft, the shaft includes a passageway and a slot leading to the passageway, and the slot has a variable width along a length of the passageway.

(16) In a further embodiment, the slot includes a first width adjacent an entrance of the passageway, the slot includes a second width along a main portion of the passageway, and the first width is greater than the second width.

(17) In a further embodiment, the passing step includes inserting the sheath into the entrance and pushing the sheath through the passageway, and, as the sheath is pushed through the passageway, the sheath is prevented from exiting the passageway via the slot and suture is permitted to exit the passageway via the slot.

(18) In a further embodiment, the passing step includes pushing the sheath through the passageway until a distal end of the sheath projects out an exit of the passageway, and the passing step further includes pulling the distal end of the sheath to remove the sheath from the passageway.

(19) In a further embodiment, after the passing step, the method includes adjusting a position of the

suture passer and then passing the sheath and the strand of suture around the bone another time before the securing step.

(20) In a further embodiment, the securing step includes securing the strand of suture to the bone by tying a knot or without tying a knot.

(21) In a further embodiment, further comprising bending a portion of a shaft based on a size of the bone before the passing step, wherein the portion is distal of an exit of a passageway.

(22) In a further embodiment, the tensioning step includes using a tensioner to tighten the strand of suture.

(23) The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a fractured bone provided with a suture wrap cerclage.

(2) FIG. 2 is a side-perspective view of an example suture passer.

(3) FIG. 3 illustrates an example assembly including a sheath and a strand of suture.

(4) FIG. 4 is a superior view of a portion of the suture passer.

(5) FIG. 5 is a side view of a portion of the suture passer.

(6) FIG. 6 is a cross-sectional view of the suture passer taken along line 6-6 in FIG. 4, and illustrates an example arrangement of a passageway and a slot adjacent an entrance of the passageway.

(7) FIG. 7 is a cross-sectional view of the suture passer taken along line 7-7 in FIG. 4, and illustrates an example arrangement of the passageway and the slot adjacent a main portion of the passageway.

(8) FIG. 8 illustrates another example assembly including a sheath and two strands of suture.

(9) FIG. 9 is a view of a suture passer and an assembly arranged relative to bone, and in particular illustrates the sheath of the assembly being inserted into the entrance of the passageway of the suture passer.

(10) FIG. 10 is another view of the suture passer and assembly arranged relative to the bone, and in particular illustrates the sheath being pushed through the passageway of the suture passer.

(11) FIG. 11 is another view of the suture passer and assembly arranged relative to the bone. In FIG. 11, the sheath has exited the passageway and the strand of suture has been passed around the bone.

(12) FIG. 12 is a side view of another embodiment of the suture passer in which a shaft of the suture passer includes a bendable section.

(13) FIG. 13 is a side view of the suture passer of FIG. 12 arranged relative to a bone. In FIG. 13, the bendable section is bent relative to its position in FIG. 12.

DETAILED DESCRIPTION

(14) This disclosure details a system and method for providing a bone cerclage. The bone cerclage maintains the relative position of fractured bone fragments to promote healing. In this disclosure, the bone cerclage is provided at least partially by suture wrapped around bone at least once. In this respect, the bone cerclage may be referred to as a suture wrap cerclage.

(15) FIG. 1 illustrates a non-limiting example of a suture wrap cerclage 20 provided relative to a bone 22 with a fracture 24. In this example, the bone 22 is a femur. Femurs can fracture, for example, from a traumatic injury or as the result of an orthopedic procedure, such as a hip replacement. The suture wrap cerclage 20 fixes the fractured fragments of the bone 22 relative to

one another, which promotes healing. While specific reference is made to femurs, it should be understood that this disclosure is not limited to femoral fractures, and extends to other bones, including but not limited to the humerus and the sternum. Further, this disclosure is not limited to being used relative to fractured bones and can be used to promote healing of cut bones.

(16) In one non-limiting example of a disclosed method, a user such as a surgeon, for example, may make use of a collection of components, referred to herein as a “system.” Among other things, an example system includes a suture passer **26** (FIG. 2) and an assembly **28** (FIG. 3) including a strand of suture **30** and a sheath **32**.

(17) The suture passer **26** is a device, specifically a medical device, configured to facilitate maneuvering of the assembly **28** around a bone in order to pass the strand of suture **30** around the bone such that a suture wrap cerclage can be provided adjacent a fracture, for example. With specific reference to FIG. 2, the suture passer **26** includes a handle **34** and a shaft **36** projecting distally from the handle **34**. The terms “distal” and “proximal” are labeled in some figures for ease of reference.

(18) The handle **34** is configured to be grasped by a user, namely a surgeon, such that the user can maneuver the shaft **36** into a particular position. In one example, the shaft **36** is made of a metallic material and is a one-piece structure. The shaft **36** has an overall length $L_{sub.1}$ between a proximal end **38** adjacent the handle **34** and a distal end **40**, which is a distal-most end of the suture passer **26** in this example. The distal end **40** may be sharp or blunt. The shaft **36** includes a straight section **42** along a length $L_{sub.2}$ between the proximal end **38** and a point **44** between the proximal and distal ends **38**, **40**. Along the straight section **42**, the shaft **36** is centered around the central longitudinal axis A of the suture passer **26**.

(19) Distal of the straight section **42**, the shaft **36** includes a curved section **46**. The curved section **46** extends from the point **44** to the distal end **40**. The curved section **46**, in this example, is curved so as to follow a constant radius R having an origin O in an expected bone location **48**. The origin O is spaced distally of the point **44**. Beginning at the point **44**, the curved section **46** initially follows the radius R by gradually extending in an inferior direction, which is a downward direction in FIG. 2, relative to the central longitudinal axis A, until the curved section **46** reaches a point directly inferior to the origin O where the radius R is perpendicular to the central longitudinal axis A. Then, following the radius R, the curved section **46** gradually extends in a superior direction opposite the inferior direction toward the central longitudinal axis A as the curved section approaches the distal end **40**. The radius R in this example corresponds to the expected contour of a bone. Suture passers having differently-sized or shaped curved sections may be available to the user such that an appropriate suture passer may be selected for a particular bone size or procedure. The curved section **46** could exhibit a different shape and need not follow a constant radius.

(20) In order to facilitate the passing of suture, the shaft **36** includes a passageway **50** and a slot **52** leading to the passageway **50**. In general, the shaft **36** is partially hollow and the hollow interior of the shaft **36** provides the passageway **50**. Further, the slot **52** is provided by a gap in a superior surface of the shaft **36**, in this example. The slot **52** extends through the superior surface of the shaft **36** to provide access to and from the passageway **50** via the slot **52**.

(21) With reference to FIGS. 4 and 5, the passageway **50** extends internally through the shaft **36** along a length from an entrance **54** adjacent the point **44** to an exit **56** adjacent the distal end **40**. Between the entrance **54** and the exit **56**, the passageway **50** extends along a main portion **58** which is bound on all sides by the shaft **36**, with the exception of the slot **52**, which, again, is provided by a gap in a superior surface of the shaft **36**. The main portion **58** is the majority of the passageway **50** by length in this example.

(22) The passageway **50** follows the contour of the shaft **36**. In particular, within the curved section **46**, the main portion **58** of the passageway **50** is curved in the same manner as the curved section **46**, as shown in FIG. 5, which illustrates the passageway **50** in phantom. Further, adjacent the entrance **54**, the passageway **50** may include an inferior wall **60** which gradually increases in depth

to avoid sharp corners within the passageway 50.

(23) The slot 52 has a variable width along the length of the passageway 50. With reference to FIG. 4, the width of the slot 52 is measured in a side-to-side direction, such as a left-to-right direction, which is perpendicular to the central longitudinal axis A. In particular, with reference to FIGS. 4, 6, and 7, the slot 52 exhibits a first width $W_{sub.1}$ adjacent the entrance 54 and a second width $W_{sub.2}$ along the main portion 58 of the passageway 50. The first width $W_{sub.1}$ is greater than the second width $W_{sub.2}$. In particular, the first width $W_{sub.1}$ is sized so permit the sheath 32 and the strand of suture 30 to enter the passageway 50 through the slot 52 adjacent the entrance 54. The second width $W_{sub.2}$ is sized so as to prevent the sheath 32 from exiting the passageway 50 via the slot 52 but to permit the strand of suture 30 to exit the passageway 50 via the slot 52. In an example, the slot 52 exhibits the second width $W_{sub.2}$ along the entirety of the main portion 58, and in particular along the entirety of the curved section 46 of the shaft 32 and at the distal end 40. The second width $W_{sub.2}$ may gradually widen adjacent the distal end 40 in other examples. Adjacent the entrance 54, the slot 52 may gradually taper from the first width $W_{sub.1}$ to the second width $W_{sub.2}$.

(24) The passageway 50 exhibits a diameter $D_{sub.1}$ along substantially the entire length of the passageway 50. The diameter $D_{sub.1}$ corresponds to a diameter $D_{sub.2}$ (FIG. 3) of the sheath 32, and in particular the diameter $D_{sub.1}$ is sized slightly larger than the diameter $D_{sub.2}$ so as to allow the sheath 32 to slide within the passageway 50 from the entrance 54 to the exit 56.

(25) With reference to FIG. 3, the strand of suture 30 is a high strength braided suture, such as Arthrex, Inc.'s FiberTape™. The strand of suture 30 may include a multifilament cover formed of a plurality of braided fibers of ultrahigh molecular weight polyethylene (UHMWPE) and fibers of polyester. The cover may surround a core formed of twisted fibers of ultrahigh molecular weight polyethylene (UHMWPE). This disclosure is not limited to any particular type of suture, however.

(26) Further, the strand of suture 30 exhibits a diameter $D_{sub.3}$ less than the diameter $D_{sub.2}$ of the sheath 32 and less than the width $W_{sub.2}$ so as to permit the strand of suture 30 to exit the passageway 50 via the slot 52. The strand of suture 30 projects from the sheath 32 by a length $L_{sub.3}$, which is greater than a length $L_{sub.4}$ of the sheath 32. The length $L_{sub.4}$ of the sheath 32 is greater than a length of the passageway 50 such that, when inserted into the passageway 50, a distal end 62 of the sheath 32 can project out of the exit 56 while a proximal end 64 of the sheath 32 is proximal to the entrance 54, as shown in FIG. 10, which increases the ease of passing the sheath 32 through the passageway 50. The length $L_{sub.3}$ of the strand of suture 30 is such that the strand of suture 30 can be passed around a bone a plurality of times and manipulated in a desired manner, such as being tied into a knot. In this respect, the length $L_{sub.3}$ may be many multiples of the length $L_{sub.4}$, and FIG. 3 is not to scale in this regard.

(27) The sheath 32, in this example, is more rigid than the strand of suture 30. In this way, a user can maneuver the sheath 32 more easily than the suture 30. In particular, it is easier for a user to insert the sheath 32 into the entrance 54 of the passageway 50 than the strand of suture 30. Further, it is easier for a user to retrieve the sheath 32 from the exit 56 of the passageway 50 than the strand of suture 30. In an example, the sheath 32 is made of a polymer material. The material of the sheath 32 may be considered semi-rigid, such that the sheath 32 is relatively stiff when compared to the strand of suture 30, but is not inflexible such that the sheath 32 can bend to conform with the shape of the curved section 46 as the sheath 32 moves through the curved section 46. The sheath 32 may be cylindrical, and may resemble a tube. The sheath 32 may be circular in cross-section, or may resemble another shape in cross-section. The sheath 32 may be either hollow or solid. The sheath 32 can be overmolded or attached to an end of the strand of suture 30 using another attachment technique.

(28) While in FIG. 3 only one strand of suture 30 projects from the sheath 32, additional strands of suture could project from the sheath 32. FIG. 8 is representative of another assembly 28 including two strands of suture 30A, 30B projecting from the sheath 32. Each strand of suture 30A, 30B is

configured similar to the strand of suture **30**. Additional strands of suture may project from the sheath **32** in other embodiments.

(29) An example method of this disclosure will now be described with reference to FIGS. **9-11**. With reference to FIG. **9**, the suture passer **26** is positioned relative to a bone **22**, which may be a femur, sternum, or humerus, as examples, adjacent a location where a bone cerclage is to be provided. As shown in FIG. **9**, the curved section **46** of the shaft **36** is arranged such that the curved section **46** extends partially around the bone **22**.

(30) When the suture passer **26** is in position, the user passes the strand of suture **30** around the bone **22** by first inserting the sheath **32** into the entrance **54** of the passageway **50**, as shown in FIG. **9**. The user continues feeding the sheath **32** into the passageway **50** by pushing on the sheath **32**, as shown in FIG. **10**. In FIG. **10**, a distal end **62** of the sheath **32** projects out of the passageway **50** and, in particular, projects from the distal end **40** of the suture passer **26**. The user can then grasp, using a tool for example, the distal end **62** of the sheath **32** and pull the sheath **32** out of the passageway **50**. In FIG. **11**, the sheath **32** is fully removed from the passageway **50** and the strand of suture **30** has been passed around the bone **22**. Next, the strand of suture **30** exits the passageway **50** via the slot **52**. The strand of suture **30** can then be secured relative to the bone **22** or can be passed around the bone **22** one or more additional times.

(31) If passing the strand of suture around the bone **22** again, the user may first reposition the suture passer **26**. Because of the strand of suture **30** has exited the passageway **50** via the slot **52**, the user does not need to fully remove the suture passer **26** from a surgical cavity to reposition the suture passer **26**. Rather, in an example, the user can reposition the suture passer **26** by moving the suture passer **26** along a length of the bone **22**, such as by sliding the suture passer **26** relative to the bone **22**. Once repositioned, the user can pass the strand of suture **30** around the bone **22** again by repeating the aforementioned steps, namely those shown and described relative to FIGS. **9-11**.

(32) In an example method, the strand of suture **30** is passed around the bone **22** at least twice. When the strand of suture **30** has been passed around the bone **22** a desired number of times, the strand of suture is tensioned and secured to the bone to provide the suture wrap cerclage **20**. The sheath **32** may be separated from the strand of suture **30** by cutting the strand of suture **30** adjacent the sheath **32**.

(33) The system of this disclosure may include one or more additional items. One such item is a suture tensioner, which is a device that allows the user to tension the strand of suture **30** suture to an appropriate level. An example of such a device is Arthrex's AR-1529 Suture Tensioner and Tensiometer. A suture tensioner is not required in all examples, but its use may increase the overall tightness of the suture wrap cerclage **20** relative to a hand-tensioned cerclage. Its use may further increase efficiency and repeatability relative to hand-tensioning.

(34) The strand of suture **30** may be secured relative to the bone **22** by tying a knot, and without implanting any structure in the body, with the exception of the strand of suture **30**. Other aspects of this disclosure, however, are "knotless," meaning the securing step includes securing the strand of suture **30** to the bone **22** without tying any knots. In an example, the strand of suture **30** is secured to the bone **22** using a knotless suture anchor, such as Arthrex's SwiveLock® or PushLock®. Various other structures, such as bone screws or bone plates may be used alone or in combination to secure the strand of suture **30** relative to the bone **22**.

(35) Another example embodiment of the suture passer **26** is illustrated relative to FIGS. **12** and **13**. In this example, the suture passer **26** is configured similar to the above-described embodiments, with some exceptions, which will now be described. In the embodiment of FIGS. **12** and **13**, the suture passer **26** is configured such that a portion of the curved section **46** of the shaft **36** is bendable to alter the effective size and/or shape of the curved section **46**. In this way, the curved section **46** can be bent to more closely match the size and/or shape of a particular bone, which in turn reduces the likelihood that the suture passer **26** will contact and/or interfere with tissue adjacent the bone.

(36) With reference to FIG. 12, the exit 56 of the passageway 50 is spaced-apart proximally from the distal end 40 of the shaft 36. In an example, the exit 56 is spaced-apart from the distal end 40 by an arc length of about 1 inch along the curved section 46. A section 66 of the shaft 36 between the exit 56 and the distal end 40 can be bent by a surgeon, for example, to adjust an effective size and/or shape of the curved section 46. Since the passageway 50 does not extend within the section 66, bending the section 66 does not impede the passage of suture 30 or the sheath 32 within the passageway 50.

(37) In an example, the section 66 is integrally formed with the remainder of the shaft 36 and is made of the same material as the remainder of the shaft 36. An exemplary material of the shaft 36 and section 66 is a metallic material such as titanium. The material of the shaft 36 and section 66 is such that the section 66 holds a new, bent shape after the section 66 is bent to that new shape. The section 66 can be bent to accommodate the size of a particular bone or section of bone, used to pass suture around that bone, and then re-bent in a different manner to accommodate a differently-sized bone or bone section. For instance, the section 66 can be bent in a first direction, used to pass suture, and then later bent in a second, opposite direction and used to pass suture relative to another bone or bone section. The shaft 36 and section 66 may be made of different materials, in other examples. In those examples, the section 66 may be formed using an additive manufacturing technique.

(38) FIG. 13 illustrates an example in which the section 66 has been bent such that the section 66 exhibits a new shape relative to its configuration in FIG. 12. The section 66 is bent by a surgeon, for example, before the shaft 36 is arranged relative to the bone 22 and before the suture 30 or sheath 32 are inserted into the passageway 50. In FIG. 13, the section 66 has been deflected proximally relative to its position in FIG. 12. Thus, the curved section 46 more closely follows the contour of the bone 22, which is relatively small in this example, than if the curved section 46 were not bent. In this way, unwanted interference between the shaft 36 and soft tissue adjacent the bone 22 is mitigated if not eliminated. The section 66 also serves to guide the sheath 32 around the bone 22 after the sheath 32 exits the exit 56, such that the sheath 32 can be readily retrieved by a surgeon.

(39) The suture wrap cerclages of this disclosure provide a reliable and effective bone cerclage, which promotes healing of fractured bones. Further, the disclosed methods and systems allow surgeons to provide suture wrap cerclages in an efficient manner without sacrificing consistency or quality.

(40) It should be understood that terms such as “proximal,” “distal,” “inferior,” and “superior” are used consistent with their art-accepted meaning. These terms should not otherwise be considered limiting.

(41) Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples. In addition, the various figures accompanying this disclosure are not necessarily to scale, and some features may be exaggerated or minimized to show certain details of a particular component or arrangement.

(42) One of ordinary skill in this art would understand that the above-described embodiments are exemplary and non-limiting. That is, modifications of this disclosure would come within the scope of the claims. Accordingly, the following claims should be studied to determine their true scope and content.

Claims

1. A system for providing a bone cerclage, comprising: a suture passer including a shaft, wherein the shaft includes a passageway and a slot leading to the passageway, wherein the passageway

extends internally to the shaft between an entrance and an exit distal to the entrance, wherein the passageway includes a main portion between the entrance and the exit, wherein the slot includes a first width adjacent the entrance and a second width along the main portion, and wherein the first width is greater than the second width; and a sheath and a strand of suture, wherein the strand of suture is attached to the sheath and projects from the sheath, wherein a length of the sheath is longer than a length of the passageway, and wherein the passageway is sized so as to allow the sheath to slide within the passageway from the entrance to the exit.

2. The system as recited in claim 1, wherein the sheath is more rigid than the strand of suture.
3. The system as recited in claim 1, wherein the sheath exhibits a larger diameter than the strand of suture.
4. The system as recited in claim 1, wherein the strand of suture is a first strand of suture and at least one additional strand of suture projects from the sheath.
5. The system as recited in claim 1, wherein: the first width is sized so as to permit the sheath and suture to enter the passageway via the slot, and the second width is sized so as to prevent the sheath from exiting the passageway via the slot and so as to permit suture to exit the passageway via the slot.
6. The system as recited in claim 1, wherein the slot is provided by a gap in a superior surface of the shaft.
7. The system as recited in claim 1, wherein: the shaft includes a curved section, and the passageway and slot are provided at least partially in the curved section.
8. The system as recited in claim 7, wherein: the suture passer includes a handle connected to the shaft, and the shaft includes a substantially straight section between the handle and the curved section.
9. The system as recited in claim 8, wherein an entrance to the passageway is provided in the substantially straight section and is spaced-apart proximally from a point at which the substantially straight section meets the curved section.
10. The system as recited in claim 1, wherein the sheath exhibits a substantially constant diameter along an entire length of the sheath.
11. The system as recited in claim 10, wherein the sheath is cylindrical.
12. The system as recited in claim 1, wherein an exit of the passageway is spaced-apart proximally from a distal end of the shaft.
13. The system as recited in claim 12, wherein the passageway does not extend within a section of the shaft between the exit of the passageway and the distal end of the shaft.
14. The system as recited in claim 13, wherein the section is made of a material different than a remainder of the shaft.
15. The system as recited in claim 1, wherein a distal end of the shaft is blunt.
16. The system as recited in claim 1, wherein an inferior wall of the passageway gradually increases in depth moving distally from the entrance such that, when the sheath comes into contact with the inferior wall, the inferior wall directs the sheath into the main portion of the passageway.
17. The system as recited in claim 1, wherein a distal end of the sheath is distal to the strand of suture.
18. The system as recited in claim 1, wherein the sheath is made of a polymer material.
19. The system as recited in claim 18, wherein the sheath is overmolded to an end of the strand of suture.
20. A system for providing a bone cerclage, comprising: a suture passer including a shaft, wherein the shaft includes a passageway and a slot leading to the passageway, wherein the slot has a variable width along a length of the passageway; and a sheath and a strand of suture, wherein the strand of suture is attached to the sheath and projects from the sheath, wherein a length of the sheath is longer than a length of the passageway, wherein an exit of the passageway is spaced-apart proximally from a distal end of the shaft, wherein the passageway does not extend within a section

of the shaft between the exit of the passageway and the distal end of the shaft, and wherein the section of the shaft between the exit of the passageway and the distal end of the shaft is bendable such that the section holds a bent shape different from an original shape of the section.
